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B. Tech CSE (ARTIFICIAL INTELLIGENCE)

22AIE315 -Natural Language Processing

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4. Problem Definition (5-7 lines):

In today's fast-paced digital environment, individuals increasingly struggle to recognize and manage their emotional well-being before it deteriorates into burnout or chronic stress. Personal journaling generates rich emotional data daily, yet this data remains largely unanalyzed as individuals cannot manually detect subtle emotional shifts or recurring behavioral patterns hidden within weeks of unstructured journal text. Current applications rely on basic sentiment analysis which fails to capture emotional complexity, temporal evolution, and mixed emotional states. This project proposes an NLP-Based Emotion-Aware Journal Analysis and Adaptive Workload Management System that processes written and spoken journal entries to extract fine-grained emotional patterns over time. The system generates personalized wellness recommendations aligned with user interests and proactively adjusts workload to prevent burnout without performing medical diagnosis or compromising user data privacy.

5. Literature survey Summary Table

Sl No	Author(s) & Proceedings(APA Style)	Year	Journal & Impact Factor	Dataset Used	Methodology/ Approach	Limitations & Future works
1	Author(s). (2025). LSTM Enhanced RoBERTa (LER): A Novel Hybrid Model for Emotion Detection in Text. Scientific Reports, Nature. https://www.nature.com/articles/s41598-025-31984-1	2025	Scientific Reports, Nature Portfolio IF: 3.9 (Q1, Clarivate JCR 2024)	ISEAR Emotion Dataset	Hybrid architecture feeding RoBERTa contextual embeddings into LSTM. 5-fold cross-validation; WordNet synonym augmentation. Benchmarked vs. RF, SVM, XGBoost, CNN, GRU, BiGRU, BERT, standalone RoBERTa. Achieves 88% accuracy.	Evaluated only on ISEAR; not tested on journal/mental health text. High computational cost. Synonym replacement may alter emotional nuance. No temporal multi-entry analysis.
2	Author(s). (2025). Emotion Recognition from Text and Voice: A Multimodal AI Approach. International Journal of Research and Analytical Studies in Engineering and Technology (IJRASET). https://www.ijraset.com/research-paper/emotion-recognition-from-text-and-voice	2025	IJRASET (Research / Conference Journal)	Combined text + audio emotion datasets	RoBERTa and DistilBERT for text classification; prosodic/acoustic feature extraction for speech; late fusion of both modalities for multi-class emotion output. Multimodal consistently outperforms unimodal.	Basic late fusion; no attention-based cross-modal weighting. Dataset scope not extensively detailed. Robustness to real-world background noise not evaluated.
3	Kurpicz-Briki, M., et al. (2025). AI-Based Approach to Burnout Identification from Textual Data. arXiv preprint arXiv:2601.17993. https://arxiv.org/abs/2601.17993	2025	arXiv (Preprint) (Pending peer review)	Russian YouTube user comments + ChatGPT-generated synthetic sentences	Fine-tunes RuBERT (Russian BERT) for burnout detection. Outputs continuous burnout probability scores. Combines synthetic and real-world data as training strategy.	Russian-language only; no English generalization. YouTube comments differ significantly from personal journal text. No longitudinal or time-series evaluation performed.
4	Author(s). (2025). Words to Waves: Emotion-Adaptive Music Recommendation System. arXiv preprint arXiv:2510.21724. https://arxiv.org/abs/2510.21724	2025	arXiv (Preprint) (ACM/IEEE submission)	EmoBank + Spotify Million Song Dataset	Wide and Deep Learning architecture. Deep: transformer + LoRA adapters predicting valence-arousal from text. Wide: cross-product feature interactions for user-emotion-genre associations. LoRA enables efficient fine-tuning.	EmoBank lacks extreme-emotion samples. Poor generalization on long text descriptions. Text input only; no voice-based emotion inference incorporated.
5	Author(s). (2025). Transforming IT Burnout Prevention with Centralized Work Pattern Monitoring Powered by AI. Preprints.org Manuscript 202502.0056. https://www.preprints.org/manuscript/202502.0056/v1	2025	Preprints.org (Not yet peer reviewed)	IT employee work behavior logs (hours worked, task completion rates, engagement levels)	AI-driven centralized monitoring system; ML models detect early burnout indicators from work behavior data; triggers proactive interventions including workload redistribution and personalized support.	Serious unresolved employee privacy concerns. IT sector only; limited generalizability to academic/personal contexts. Not yet peer reviewed.
S1	Author(s). (2025). Deep Emotion Recognition in Textual	2025	Artificial Intelligence Review, Springer IF:	IEMOCA P, MELD, EmoryNL P,	Systematic survey of ERC architectures: MLP, CNN, LSTM, GRU, GNN, Attention, Transformers. Reviews ACL,	Covers literature up to 2024; may miss latest 2025 developments. Focus on conversational ERC, not single-author journal text.

	Conversations: A Survey. Artificial Intelligence Review, Springer. https://link.springer.com/article/10.1007/s10462-024-11010-y		13.9 (Q1, Clarivate 2024)	DailyDialog (systematic survey review)	EMNLP, NAACL, AAAI, INTERSPEECH, IEEE TAC venues. Includes open-access categorized repository.	Multilingual ERC and computational cost benchmarking remain open gaps.
6	Author(s). (2025). Detecting Emotion Drift in Mental Health Text Using Pre-Trained Transformer Models. arXiv preprint arXiv:2512.13363v1. https://arxiv.org/html/2512.13363v1	2025	arXiv (Preprint) (Pending peer review)	Mental health text from Reddit and clinical forums	Multi-stage pipeline: sentence segmentation → per-sentence emotion labeling (DistilBERT, RoBERTa, DeBERTa-v3) → drift score computation across sentences → explainable visualization interface for interpretability.	Operates within a single document only; does not track emotional changes across multiple journal entries over time. No real-world clinical validation. Does not handle multilingual or code-switched text.
7	Author(s). (2025). A Comprehensive Review of Multimodal Emotion Recognition: Techniques, Challenges, and Future Directions. PMC / Applied Sciences, MDPI. https://pmc.ncbi.nlm.nih.gov/articles/PMC12292624/	2025	PMC / Applied Sciences, MDPI IF: 2.5 (Q2, Clarivate 2024)	Survey of benchmarks: IEMOCAP, CMU-MOSI, DEAP, AVEC, etc.	Comprehensive survey of text, speech, and vision modalities. Reviews early fusion, late fusion, and hybrid attention-based cross-modal fusion strategies. Evaluates benchmark datasets and evaluation methodologies across the field.	No novel model introduced. Breadth limits per-architecture technical depth. Some reviewed architectures may already be superseded by newer 2025 work.
8	Author(s). (2025). Passive AI Detection of Stress and Burnout Among Frontline Workers. Nursing Reports, PMC/MDPI. https://pmc.ncbi.nlm.nih.gov/articles/PMC12655262/	2025	Nursing Reports, PMC/MDPI IF: 2.5 (Q2, Clarivate 2024)	10 peer-reviewed studies via PubMed, IEEE Xplore, Scopus, ACM, Web of Science	Narrative review using Mixed Methods Appraisal Tool. Evaluates passive AI using wearable biometric sensors, EHR interaction logs, and communication pattern metrics. Reports 75–90% classification accuracy across included studies.	Narrative (not systematic) review; potential selection bias. Heavy focus on physiological data, not language-based signals. Small sample sizes; limited cross-cultural diversity.
9	Author(s). (2025). A Continuous Music Recommendation Method Considering Emotional Change. Applied Sciences, 15(13), 7222. MDPI. https://www.mdpi.com/2076-3417/15/13/7222	2025	Applied Sciences, MDPI IF: 2.5 (Q2, Clarivate 2024)	Self-reported emotion data + music interaction logs (Thayer 2D model-based)	Classifies emotions in Thayer's valence-arousal quadrants. Multiple regression for music/user emotion vectors. Markov chains model emotional state transitions over time for continuous dynamic recommendation toward a target state.	Self-reported data introduces subjectivity bias. No long-term real-world validation conducted. Combined regression and Markov modeling increases computational complexity.
10	Author(s). (2024). Implementing AI-Driven Employee Wellbeing Solutions: A Data-Centric Approach to Preventing Burnout. ResearchGate Publication 387089926. https://www.researchgate.net/publication/387089926	2024	ResearchGate (Organizational Behavior domain)	Employee behavioral and performance data (organizational context)	Structured data-centric framework for AI wellness deployment. Analyzes behavioral/performance data for burnout risk indicators. Establishes continuous feedback loops with personalized recommendations integrated into workflow management layer.	Primarily conceptual; limited empirical validation in real deployments. Enterprise context may not generalize to individual-user personal wellness systems. Privacy/ethics concerns acknowledged but not fully resolved.
S2	Author(s). (2025). A Survey on Multimodal Music Emotion Recognition.	2025	arXiv / ACM Computing Surveys (Submitted;	Survey across MER benchmark	Systematic survey of multimodal MER: audio spectrograms, lyrics/NLP-text (BERT, RoBERTa), metadata. Reviews early, late, and	No novel model introduced. Limited physiological signal coverage. Free-form journal text inference (vs. structured lyrics)

	arXiv preprint arXiv:2504.18799v1. (Submitted to ACM Computing Surveys). https://arxiv.org/html/2504.18799v1		ACM CS IF: 23.8, Q1)	ks: EmoBank, DEAP, IAPS, Million Song Dataset, etc.	attention-based intermediate fusion. Maps full NLP-to-music recommendation pipeline including user emotion inference to personalized output.	identified as underexplored gap. Cold-start challenges for new users acknowledged but not resolved.
11	Author(s). (2025). Emotion Detection Using Transformer Model With Deep Learning. SSRN Electronic Journal. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5198476	2025	SSRN Electronic Journal (Preprint repository)	Mental health- adjacent text datasets	Benchmarks BERT and RoBERTa across multi-class emotion classification. Tests multiple transformer configurations vs. traditional ML baselines. Optimized configurations achieve >90% accuracy on mental health-adjacent text.	Dataset and full methodology not comprehensively detailed in preprint; limits reproducibility. Mental health text may not represent informal journal writing style. No longitudinal or temporal analysis explored.
12	Author(s). (2025). Speech Emotion Recognition in Mental Health: Systematic Review of Voice-Based Applications. JMIR Mental Health, 12, e74260. https://mental.jmir.org/2025/1/e74260	2025	JMIR Mental Health IF: 5.8 (Q1, Clarivate 2025)	Systematic review of SER studies in clinical and mental health contexts	Systematic review covering prosodic, spectral, and linguistic feature extraction; deep learning SER architectures; available datasets; clinical use-case evaluations. Explicitly evaluates ethical deployment considerations for wellness contexts.	No new SER model proposed. Heterogeneity across included studies limits direct performance comparison. Lab datasets may not generalize to naturalistic everyday voice journal recordings.
13	Kurpicz-Briki, M., & Neff, B. (2024). Using NLP to Find Indication for Burnout with Text Classification: From Online Data to Real- World Data. arXiv preprint arXiv:2409.14357. https://arxiv.org/abs/2409.14357	2024	arXiv (Preprint) (Pending peer review)	Anonymo us real- world free-text responses with OBI scores + BurnoutE xpressions dataset (real + GPT-3.5 synthetic)	Trains GermanBERT-based classifier on online data; evaluates on real-world workplace dataset to measure domain shift. Creates BurnoutExpressions dataset. Interdisciplinary focus group provides qualitative validation.	German-language only; no English generalization. Real- world dataset size is small. GPT- 3.5 synthetic data may not capture authentic burnout language. No longitudinal or time-series evaluation.
14	Author(s). (2025). Emotion-Driven Music Recommendation System Based on Fully Convolutional Recurrent Attention Networks. Alexandria Engineering Journal (ScienceDirect). https://www.sciencedirect.com/science/article/pii/S1110016825004223	2025	Alexandria Engineering Journal, Elsevier (ScienceDirect) IF: 6.2 (Q1, Clarivate 2024)	Standard music emotion recognitio n benchmar ks	Fully convolutional recurrent attention network jointly models audio-based music features and user emotional state vectors. Attention mechanism aligns music features with detected user emotion for emotionally coherent recommendations.	Focuses on audio features only; no user text or voice input for emotion inference. Computationally expensive architecture. Evaluation dataset may not reflect culturally diverse music preferences.
15	Meduri, R., et al. (2025). Human- Centered AI for Personalized Workload Management: A Multimodal Approach to Preventing Employee Burnout. Journal of Information Processing and	2025	Journal of Information Processing & Digital Communication (Enpress) (Peer- reviewed journal)	Survey of 320 participant s from healthcare and IT sectors	Quantitative survey measuring 4 AI integration dimensions: integration intensity, training adequacy, personalization level, feedback frequency. Statistical analysis correlates dimensions with self- reported burnout levels across participants.	Self-reported burnout measures introduce common method bias. Sample limited to healthcare and IT; limits generalizability to students or personal wellness users. Findings are correlational, not causal.

	Digital Communication. https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/6918/3799					
S3	Baliyan, P., & Dahiya, S. (2024). A Review on Emotion Detection by Using Deep Learning Techniques. Artificial Intelligence Review, Springer. https://doi.org/10.1007/s10462-024-10831-1	2025	Artificial Intelligence Review, Springer IF: 13.9 (Q1, Clarivate 2024)	Systematic review across 330 research papers (2013–2023) from IEEE, ScienceDirect, Springer, Scopus, Web of Science	Systematic literature review of text-based emotion detection (2013–2023). Reviews ML classifiers (SVM, RF, LR), deep learning (CNN, LSTM, BiGRU, BERT, RoBERTa), emotion models (Ekman, Plutchik, Russell), datasets (ISEAR, SemEval, GoEmotions), and applications in mental health, education, and social media.	Covers literature up to 2023; may miss latest 2024–2025 LLM-based developments. Primarily focuses on textual modality; limited coverage of multimodal and voice-based emotion detection. Cross-linguistic and low-resource language emotion detection remain open gaps.

6. Identified Research Gaps:

1. Absence of Longitudinal Journal-Specific Emotion Analysis

The majority of NLP emotion detection models reviewed, including LER (Paper 1) and the SSRN transformer benchmarking paper (Paper 11), evaluate on controlled single-instance text datasets such as ISEAR and do not assess emotional trends across multiple entries over time. The emotion drift paper (Paper 6, arXiv 2025) introduces intra-text drift but limits its scope to within a single document. No existing system applies temporal emotional trend analysis specifically to daily personal journal entries, which have a fundamentally different structure, vocabulary, and emotional rhythm than social media text, clinical interview transcripts, or online forum posts.

2. Absence of True Multimodal Input in Journal Wellness Systems

While multimodal emotion recognition has advanced significantly as demonstrated by the PMC comprehensive review (Paper 7) and the IJRASET voice-text paper (Paper 2), no existing journaling or personal wellness system integrates both text and voice modalities simultaneously for emotion analysis within a unified pipeline. Speech emotion recognition has been validated for mental health contexts by the JMIR systematic review (Paper 12), yet it has not been combined with NLP-based journal text analysis in a single personal wellness application. This creates a clear gap that this project fills by designing a system that processes both written and spoken journal entries and fuses their emotional signals for richer detection.

3. Static Emotion-to-Recommendation Mapping Without User Personalization

Existing emotion-based music recommendation systems including the continuous Markov-chain model (Paper 9, MDPI) and the convolutional attention network (Paper 14, ScienceDirect) map detected emotions to music recommendations but do not personalize these based on individual user interests collected at onboarding. Recommendations remain generic to emotional categories rather than being tailored to what a specific user enjoys. Furthermore, none of the reviewed recommendation systems incorporate voice-based emotion inference as an input to the recommendation pipeline, relying exclusively on text or self-report.

4. Disconnection Between Emotional Insights and Workload Action

Burnout detection systems reviewed including Kurpicz-Briki et al. (Papers 3 and 13) and the passive monitoring review (Paper 8) successfully identify burnout risk from textual or behavioral signals but stop at detection. They do not translate detected emotional states or burnout risk scores into actionable, real-time workload adjustments. Similarly, adaptive workload frameworks (Papers 5 and 15) focus on organizational-level behavioral data rather than personal emotional state derived from journal content. No existing system creates a closed-loop pipeline from journal-based emotion detection through burnout risk scoring directly to personalized task reprioritization and rest recommendations.

5. Lack of Privacy-Aware, Ethically Designed Personal Wellness Agents

The continuous workplace monitoring system (Paper 5, Preprints.org) raises significant unresolved privacy concerns. The JMIR SER review (Paper 12) explicitly identifies ethical deployment as an open challenge. Most existing burnout and stress detection systems require access to sensitive organizational data, email logs, or physiological sensors all involving significant privacy trade-offs. No existing system achieves the combination of journal-based emotion analysis, multimodal input, personalized recommendation, and adaptive workload management within a design that explicitly limits data exposure to derived emotional features only, as proposed in this system.

7. Novelty / Contribution of the Proposed Work

1: Multimodal Journal Input - Text and Voice Fusion

The system is the first to integrate both written text and voice recordings as journal input modalities within a personal wellness context. Voice entries undergo dual processing: speech-to-text transcription and acoustic tone analysis (pitch, speech rate, vocal intensity). Emotional signals from both modalities are fused to produce a richer and more accurate emotional representation than either modality alone could provide.

2: Longitudinal Emotional Trend Analysis with Burnout Risk Scoring

A proprietary Emotional Momentum Score is computed per user, combining emotion intensity, duration, and frequency across journal entries over days and weeks. Moving average computation and z-score anomaly detection identify sustained stress trends and escalating burnout risk going beyond single-entry classification to genuine temporal pattern recognition specific to personal journal data.

3: Interest-Profiled Personalized Wellness Recommendations

At onboarding, users provide their interests including music genres, physical activities, and hobbies. The system continuously refines this profile from recurring themes in journal content. Wellness recommendations music categories, physical activities, journaling prompts are matched to the specific user's interests and current emotional state, replacing generic category-level suggestions with personalized, contextually appropriate support.

4: Closed-Loop Adaptive Workload Management Agent

An intelligent agent monitors the Emotional Momentum Score and proactively intervenes by generating daily energy assessments, suggesting task reprioritization, and recommending rest intervals. This creates a closed-loop pipeline from journal input to actionable behavioral support - transforming the system from passive insight generation into active, sustainability-focused intervention.

5: Privacy-Preserving Ethical Architecture

The system limits analysis to derived emotional features only, raw journal text is never exposed in the reporting or recommendation pipeline. No medical diagnosis or therapeutic claims are made. This ethical-by-design approach makes the system safe for academic, professional, and personal wellness deployment, directly addressing the critical privacy gap identified across reviewed literature.

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